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EP1108 - Modern Physics - Exam 2

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Two protons (with rest mass m) moving towards each other with equal and opposite velocity (v), collide and produce two protons and another particle of rest mass M. What is energy of each proton (before the collision) if all the reactant particles are produced at rest in the C.O.M frame after the collision? (Assume c=1) [2 pts]

- [ ] 2m + M
- [x] m+ M/2
- [ ] M
- [ ] m+2M+(M^2/2m)

Two nuclei A and B have atomic mass numbers 2 and 128 respectively. The ratio of their radii (radius of nucleus A over that of nucleus B) is equal to [1 pt]

- [x] 1:4
- [ ] 1:8
- [ ] 1:64
- [ ] None of the above

A particle of rest mass m has (relativistic) K.E. equal to 3mc^2. So total momentum of the particle is equal to? [1 pt]

- [ ] sqrt(6) mc
- [ ] 2 mc
- [x] sqrt(15) mc
- [ ] sqrt(3) mc

One gm of radium (mass number 226) has an activity equal to 1 curie. What is the approximate half-life of Radium in years? (Avogadro number = 6 x 10^[23]) [2 pts]

- [ ] 160
- [x] 1600

- [ ] 1.6
- [ ] 0.16

The Stern-Gerlach experiment helped prove which of the following statements [1 pt]

- [x] Spin angular momentum is quantized
- [ ] Orbital angular momentum is quantized
- [ ] Both spin and orbital angular momentum are quantized
- [ ] None of the above

Two nuclei X and Y have mass numbers equal to 23 and 25 and atomic number equal to 12 and 14, respectively. These nuclei can be considered as : [1 pt]

- [ ] Isomers
- [ ] Isobars
- [ ] Isotopes
- [x] Isotones

Which mathematical quantity is a probe of spherical symmetry of the charge distribution of nucleus? [1 pt]

- [ ] Dipole moment
- [x] Quadrupole moment
- [ ] Octupole moment
- [ ] None of the above

Which of the following statements are true?[1pt]

- [ ] Particles with spin (1/2)hbar obey Bose-Einstein statistics
- [x] Photons obey Bose-Einstein statistics

- [x] In the classical limit, particles obey Maxwell-Boltzmann distribution
- [x] Electrons obey Fermi-Dirac statistics

If the kinetic energy of two particles, K_a and K_b are 900 Joules and 400 Joules respectively, what would be the ratio of their de-Broglie wavelengths, λ_a/λ_b? [1 pt]

- [ ] 400/900
- [ ] 900/400
- [x] 20/30
- [ ] 30/20

Which of the following can be valid wave functions? Choose all that are applicable. [1 pt]

- [ ] x*tan(x)
- [ ] x + 2cos(x)
- [x] √x * sin(x)
- [ ] sec(x)+cot(x)

What is the statistical postulate?[1pt]

- [x] All microstates in an ensemble are equally probable
- [ ] All microstates in a micro-canonical ensemble are equally probable
- [ ] All microstates in a canonical ensemble are equally probable
- [ ] All microstates in a grand-canonical ensemble are equally probable

Calculate the wavelength of light that has its third minimum at an angle of 30º when falling on double slits separated by 3.00 μm [2 pt]

- [ ] 400 nm

- [ ] 500 nm
- [x] 600 nm
- [ ] 700 nm

What is the ground state energy of an electron confined within a box of 1 Å? The mass of an electron is 9.1 x 10^-31 kg. [1 pt]

- [ ] 6.016 x 10^-20 J
- [ ] 2.016 x 10^-18 J
- [ ] 5.023 x 10^-18 J
- [x] 6.016 x 10^-18 J

Which of the following is the eigenfunction of the momentum operator with the positive eigenvalue ħk? [2 pt]

- [ ] e-ikx (exp (-ikx))
- [x] e+ikx (exp (+ikx))
- [ ] cos (kx)
- [ ] sin (kx)

A 1 MHz plane-wave is traveling in a dispersive medium has a plane velocity of 300 x 10^6 m/s. The phase velocity as a function of wavelength λ is given by $V_p = K\sqrt{\lambda}$. What is the group velocity of the wave? [2 pt]

- [x] 150 x 10^6 m/s
- [ ] 300 x 10^6 m/s
- [ ] 600 x 10^6 m/s
- [ ] 75 x 10^6 m/s

The Quantum mechanical tunneling effect is notably observed in the case of [1 pt]

- [ ] X-rays
- [ ] Gamma-rays
- [x] Alpha particles
- [ ] Beta particles

Consider a micro-canonical ensemble of bosons confined in an infinite potential quantum box. The energy of the ground state in the box is 1eV and there are 2 bosons inside the box with a total energy of 50eV. How many microstates are present in this ensemble? [2pt]

- [ ] 1
- [x] 2
- [ ] 3
- [ ] 4

When a substance undergoes a second order phase transition, which of the following is true?[1pt]

- [x] Its free energy changes continuously
- [ ] Its free energy changes discontinuously
- [ ] Its entropy changes continuously
- [x] Its entropy changes discontinuously

[2pt]

Consider a laser system in which the energy gap between the 2 lasing levels is 0.1eV and the lifetime of the upper level is $10^{-5}s$. The refractive index of the lasing medium is $n_0 = 1.5$. What are the Einstein A and B coefficients
a) $A = 10^5 s^{-1}$ and $B = 1.4 \times 10^{21} J^{-1} s^{-2} m^3$
b) $A = 10^4 s^{-1}$ and $B = 3.5 \times 10^{20} J^{-1} s^{-2} m^3$
c) $A = 10^5 s^{-1}$ and $B = 1.5 \times 10^{23} J^{-1} s^{-2} m^3$
d) $A = 10^5 s^{-1}$ and $B = 4.3 \times 10^{22} J^{-1} s^{-2} m^3$

- [ ] a

- [ ] b
- [ ] c
- [x] d

[3pt]

In a canonical ensemble, the energy of the microstates is allowed to fluctuate keeping the temperature constant. We have shown in class that the average energy is $\langle E \rangle = -(\partial(\ln Q)/\partial\beta)|_{N,V}$. Now the fluctuation in energy is defined as $\Delta E = E - \langle E \rangle$. Which of the expressions below give the correct average energy fluctuation square, $\langle (\Delta E)^2 \rangle$?

a) $\langle (\Delta E)^2 \rangle = \langle E^2 \rangle - 2\langle E \rangle^2$

b) $\langle (\Delta E)^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2$

c) $\langle (\Delta E)^2 \rangle = \frac{1}{Q}\frac{\partial^2 Q}{\partial \beta^2}|_{N,V} - (\frac{\partial \ln Q}{\partial \beta})^2|_{N,V}$

d) $\langle (\Delta E)^2 \rangle = \frac{\partial^2 \ln Q}{\partial \beta^2}|_{N,V} - (\frac{\partial \ln Q}{\partial \beta})^2|_{N,V}$

- [ ] a
- [x] b
- [x] c
- [ ] d

A system has 3 quantum states which have energies 0.45eV, 0.5eV, and 0.6eV. It is kept at a temperature of 1000K and a large bath of electrons is available to fill up the quantum states. The chemical potential is 0.4eV. What is the average number of electrons present in this system? [2pt]

- [ ] 1.34
- [ ] 0.59
- [x] 0.69
- [ ] 0.81

Planck's law of radiation is an example of [1pt]

- [ ] Maxwell-Boltzmann distribution
- [ ] Fermi-Dirac distribution
- [x] Bose-Einstein distribution
- [ ] Canonical partition function

What is the Gibbs free energy due to surface tension of a hollow water bubble with radius 1cm? (surface tension of water is 0.072N/m) [2pt]

- [ ] 2 X 10^-5J
- [ ] 10^-5J
- [ ] 9 X 10^-5J
- [x] 1.8 X 10^-4J

[2pt]

The canonical partition function of a gas of ultra-relativistic particles is

$$Q(N,\beta,V) = \frac{1}{N!}\left[8\pi V\left(\frac{1}{\beta hc}\right)^3\right]^N \tag{1}$$

where the symbols have their usual meaning. What is the average energy per particle?

- [x] 3kBT
- [ ] (3/2)kBT
- [ ] (1/2)hc
- [ ] hc

[1pt]

Consider a system which has a free energy given by

$$G = \begin{cases} A(T-T_0)^2 \text{ for } T \leq T_0 \\ B(T-T_0)^3 \text{ for } T > T_0 \end{cases}$$

where A and B are non-zero constants and T is temperature. Which of the following statements are true?

- [ ] There is no phase transition at T=T0
- [x] The free energy changes continuously at T=T0
- [x] There is a second order phase transition at T=T0
- [ ] There is a third order phase transition at T=T0

A laser produces light at a wavelength of 2000 Angstrom inside the active medium. The medium has refractive index of 1.7. What is the energy gap of the lasing levels? [1pt]

- [ ] 6.1eV
- [x] 3.7eV
- [ ] 2.1eV
- [ ] 10.4eV

A lasing medium has refractive index 4.5 and the upper lasing level has a lifetime of 10^-9 sec. What is the maximum length of the lasing cavity so that there is only one resonant mode in laser output? [2pt]

- [x] 0.4m
- [ ] 1.0m
- [ ] 0.1m
- [ ] 2.3m

If the band gap of material A is less than the band gap of material B, the conductivity of material A at temp T will be [2 pts]

- [ ] equal to B
- [ ] less than B
- [x] more than B
- [ ] none of the above

According to Kronig-Penney model, as the energy increases, the width of successive energy bands [2 pt]

- [ ] increases
- [x] decreases
- [ ] is not affected
- [ ] none of the above

A semiconductor at 0K behaves like an insulator because it has [2 pt]

- [ ] zero band gap
- [ ] low band gap
- [x] no electron in conduction band
- [ ] partially filled valance band
- [ ] Other: _____

In a pure crystalline silicon material, the forbidden energy gap has [2 pt]

- [ ] empty energy states
- [x] no energy states
- [ ] filled energy states
- [ ] partially filled energy states

If the uppermost energy band of a solid is partially filled, it will behave like a [2 pt]

- [ ] insulator
- [ ] semiconductor
- [x] conductor
- [ ] none of the above

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